

CSCI 49900: Progress Report 1

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I. PROJECT OVERVIEW

The Language Model Lab project is an interactive platform meant to demonstrate the mechanics of Large Language Model (LLM) inference optimization and mechanistic interpretability. Please see the submitted abstract for more details.

II. INDIVIDUAL PROJECT CONTRIBUTIONS

We spent part of the first weeks discussing our long-term contributions to the project, which we list here. The details of these are subject to change as the project progresses.

A. James

James will focus on the backend and on custom CUDA quantization. Standard LLM weights are stored in 16-bit floating point numbers (FP16). James is investigating how to implement a smaller representation, such as 8-bit integer (INT8) quantization, to reduce memory bandwidth bottlenecks and increase token generation throughput. He will also build a CUDA C++ kernel to perform some of the model's operations, one possibility being General Matrix Multiply (GEMM), directly on the INT8 weights. He is currently establishing a repository structure and environment management using Docker and Conda.

B. Alex

Alex will work on the parts of the backend that will allow users to experiment with the model. Part of our project will allow a user to prompt a language model while intervening it in various ways and observing the effects these experiments have in its responses. Alex ran some basic tests on Jupyter notebooks using various models to ensure the experiments are viable, to check which experiments would be interesting to run, and to familiarize himself with the process of writing hooks, i.e., functions that intercept the model at various intermediate points before it computes its final next-token probabilities or logits.

C. Kailun

Kailun will focus on the front end of the application, which will be vital for the user to be able to learn about the model optimizations and for them to be able to interact with the model. He has been researching how to best use React to design the user interface and D3.js to create interactive graphics. He has also been designing a basic wire-frame for the website. Next week, he will keep working on developing D3.js graphics, and he will solidify a first version of the layout design for the website through meetings with other team members.

III. PROGRESS TO DATE AND FUTURE DIRECTIONS

Here we list some tasks we have addressed so far, as well as directions for the upcoming week.

A. Model Selection and Profiling

Part of the first weeks was spent selecting and studying candidate models for our application. Given our hardware constraints, the compute demand imposed by the models, and our particular interests, we agreed that models in the range of 500M to 1B parameters would work best.

James shortlisted two open-source models he is interested in working with: TinyLlama 1.1B and Qwen1.5-0.5B. Next, he will profile these baseline FP16 models.

Alex ran some small experiments to test various models in the Pythia, Qwen, and TinyLlama families within the mentioned parameter range, and compared their responses.

We expect to finalize model selection this following week.

B. General Application Design and Integration

All of us spent some time discussing the features that should be included in the application, and what these should look like.

James proposed showcasing the impact of compression and other optimizations that his custom kernel work will do to the model, perhaps compared to an unoptimized baseline. Kailou proposed creating visualizations of the internal architecture of the model for the user using 3d.js, so they can better understand what they are modifying through graphics, and how the internals of the model function. Alex proposed the user should be able to observe model responses before and after performing interventions side-by-side, with some user-friendly graphical intervention controls. All of us also discussed what should be included in the basic API contract between front end and back end.

This following week we will further refine which aspects should be included to have a minimal viable product, and what the user experience should be like. Some items to discuss include defining the basic layout for the application, what kind of graphics should be displayed, what are the most essential features for a first version, which sections the website should contain, and what kind of information these should have. We will also finalize a common repository where we can collaborate, since our work during the first weeks was exploratory and didn't require immediate integration.